



Computer Aided Medical Procedures II

Robustness Under Systemic Error

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Outline

1 Problem Definition and Overview

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1- Problem Definition and Overview:

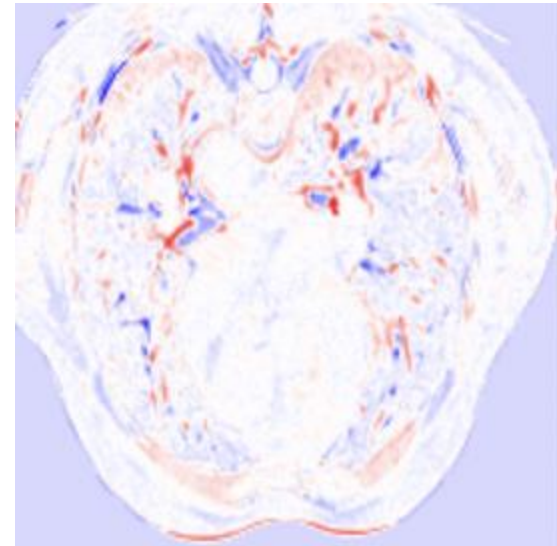
Why Robustness Matters in Registration?

Image registration

Estimate spatial correspondance to perfectly align and overlay a pair of medical images

Used in:

- longitudinal imaging
- radiation therapy
- surgical planning



But small acquisition differences can strongly affect performance

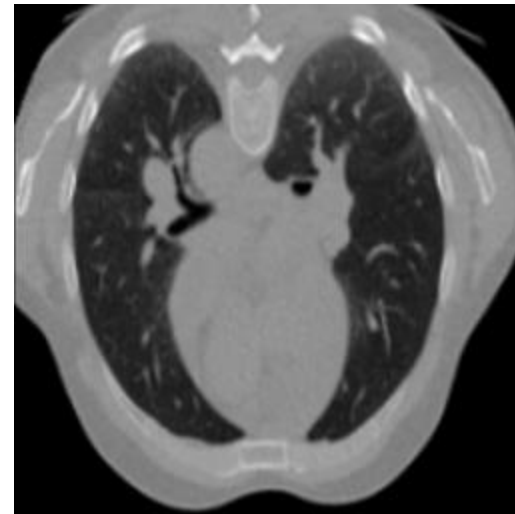
1- Problem Definition and Overview:

Exercise Pipeline



Model and Dataset

- **Dataset:** Learn2Reg LungCT
- 20 inspiration/expiratory CT pairs
- **Model:** uniGradICON foundation model, Pretrained deformable registration network model



2- Dataset & Task

1 Establish the Baseline

1. **Input:** Fixed and moving image
2. **Preprocessing:** Image resizing and normalisation
3. **Model:** Uses a multi-resolution **3D U-Net registration network**
4. **Predict:** dense deformation field (a voxel-by-voxel displacement map showing how every point should move)
5. **Similarity metric:** LNCC, mTRE, Dice score
6. **Regularization:** Helps reduce folding and impossible deformations
7. **Output:** -Registered/aligned lung CT image and a Deformation field

2- Dataset & Task

2 Corruption

We tried several different approaches to corrupt our dataset :

Z-Resolution
Downsampling

Gaussian noise

FOV Truncation

2- Dataset & Task

2 Corruption

We tried several different approaches to corrupt our dataset :

**Z-Resolution
Downsampling**

Gaussian noise

FOV Truncation

Downsampled the z-axis by a factor of 4

Simulate: Thick-slice CT acquisition, different hospital scanning protocols,
loss of axial resolution (e.g. 1 mm vs 2.5mm slices)

Effect: Destroyed fine axial structures such as lung fissures

But : registration performance slightly improved instead of degrading

2- Dataset & Task

2 Corruption

We tried several different approaches to corrupt our dataset :

Z-Resolution
Downsampling

Gaussian noise

FOV Truncation

Added random intensity noise to CT images

Simulate: Scanner Noise, Low-dose CT acquisition, reconstruction artifacts

But : artefact was less clinically meaningful than FOV truncation

2- Dataset & Task

2 Corruption

We tried several different approaches to corrupt our dataset :

Z-Resolution
Downsampling

Gaussian noise

FOV Truncation

Simulate: Different Hospitals/scanners capture different axial ranges,
Some scans systematically miss anatomy
Common in multi-site clinical datasets

We tried:

- 1) Top-slice truncation -> **no effect**
- 2) Mid-lung truncation: Zeroed slices 60-130

2- Dataset & Task

3

Used Measurements

LNCC

(Localized Normalized Cross Correlation):

Measures how similar the warped moving image is to the fixed image.

$$LNCC(I_f, I_m) = \frac{\sum (I_f - \bar{I}_f)(I_m - \bar{I}_m)}{\sqrt{\sum (I_f - \bar{I}_f)^2 \sum (I_m - \bar{I}_m)^2}}$$

Where:

- I_f = fixed image
- I_m = warped moving image
- $\bar{I}$ = local mean intensity

Folded Voxels (Jacobian Folding)

Measures whether the deformation field creates physically impossible folding or flipping of anatomy.

$$\text{Flips} = \sum_x \mathbf{1}(\det(J_\phi(x)) < 0)$$

Where:

- $\phi(x)$ = deformation field
- $J_\phi(x)$ = Jacobian matrix
- $\det(J_\phi(x)) < 0$ = folding occurs
- $\mathbf{1}$ = counts voxels where folding happens

2- Dataset & Task

4 Fix the corrupted data sets

Gaussian noise:

FOV Truncation

2- Dataset & Task

4 Fix the corrupted data sets

Gaussian noise:

FOV Truncation

Challenge: High-frequency intensity distortions mask features

Solution: Gaussian Smoothing Filter

- Applies a spatial blurring kernel to suppress pixel-level noise
- **Trade-off:** Successfully cleans the signal for stable model gradient calculation, but slightly reduces sharp edge definition

2- Dataset & Task

4 Fix the corrupted data sets

Gaussian noise:

FOV Truncation

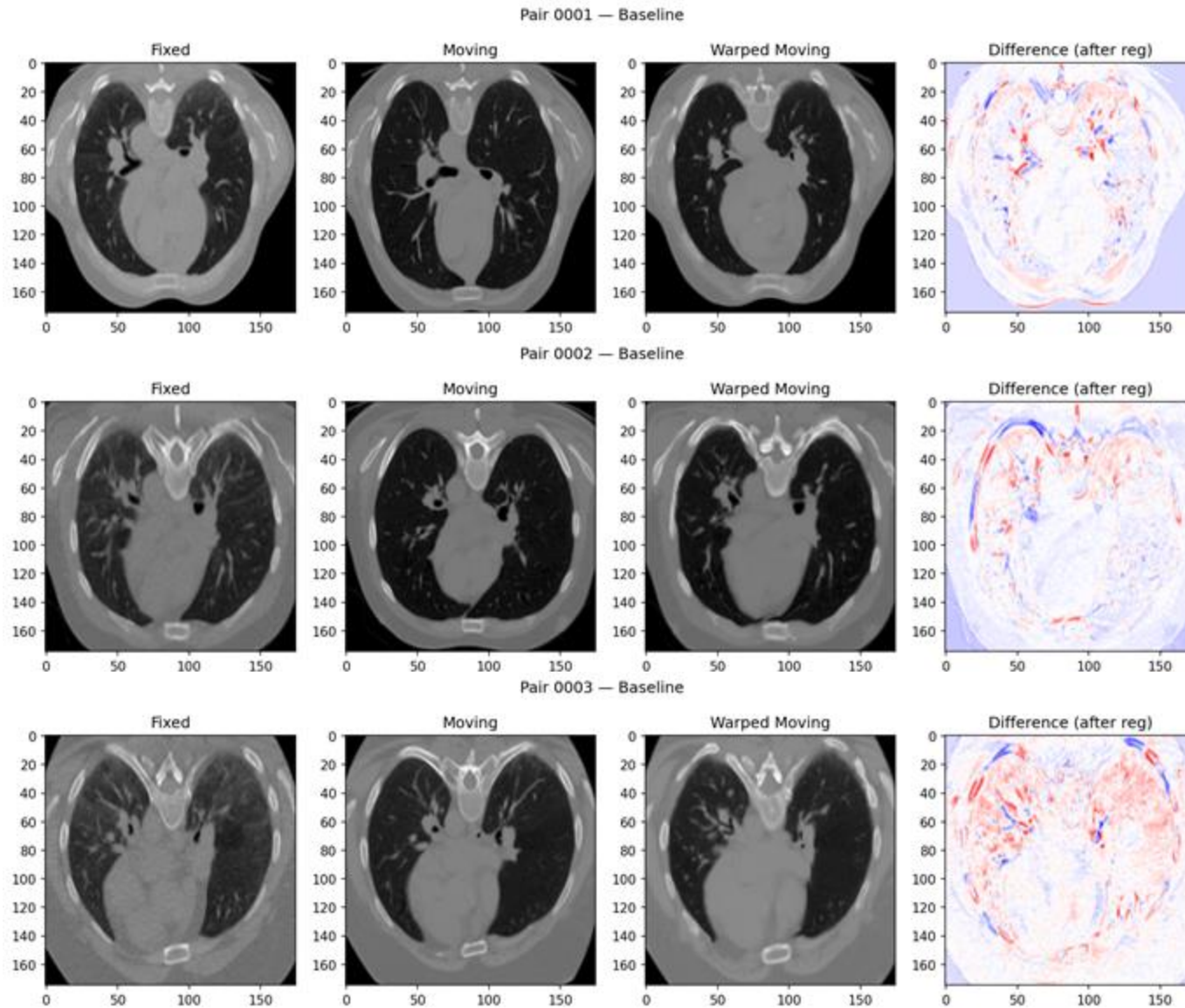
Challenge: Lost gradients and landmarks

Approach A: Linear Boundary Interpolation

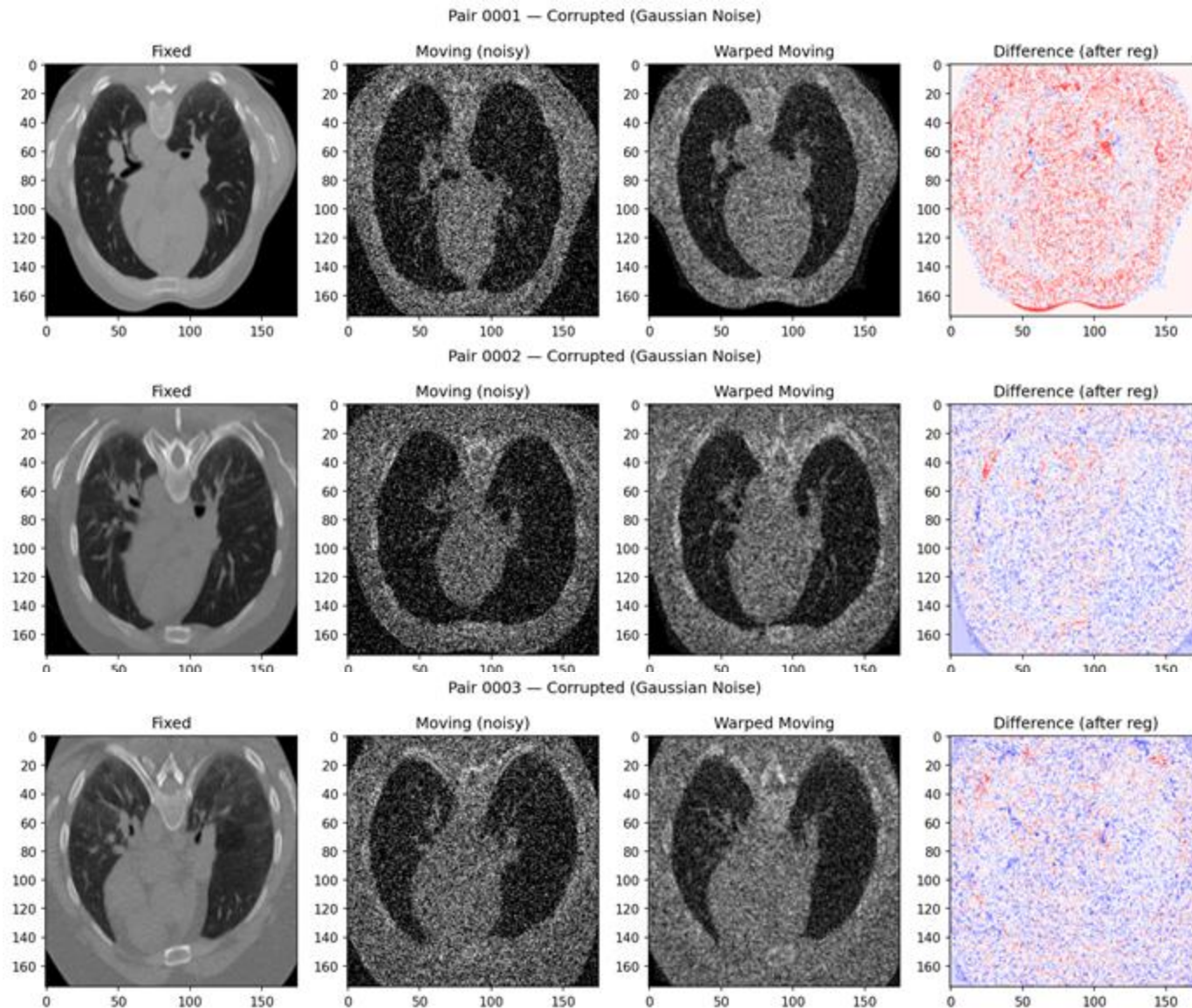
- Smooths edges but fails to recover missing anatomical structures

Approach B: Reference-based Inpainting (Our Fix)

- Uses corresponding slices from the healthy **Fixed Image** as a template to fill the gap
- **Effect:** Restores structural continuity; reduces folded voxels (Flips) back to baseline levels

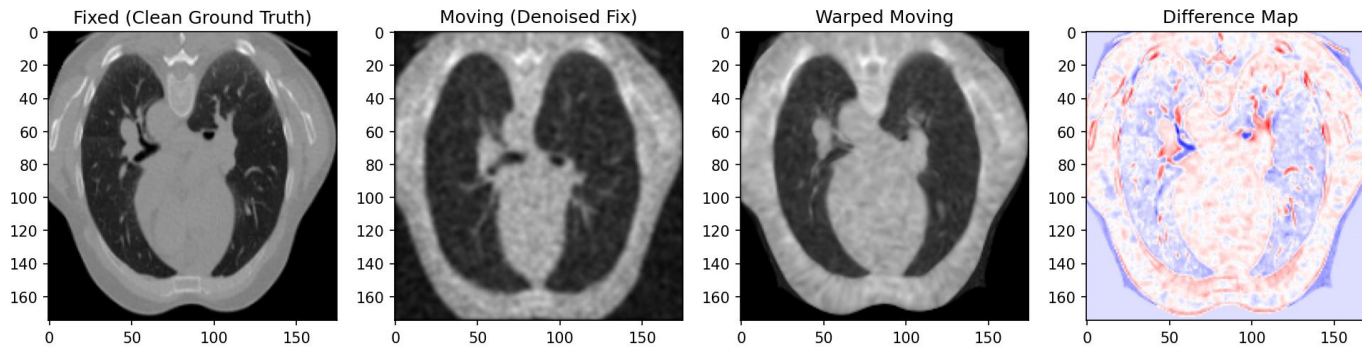


Baseline

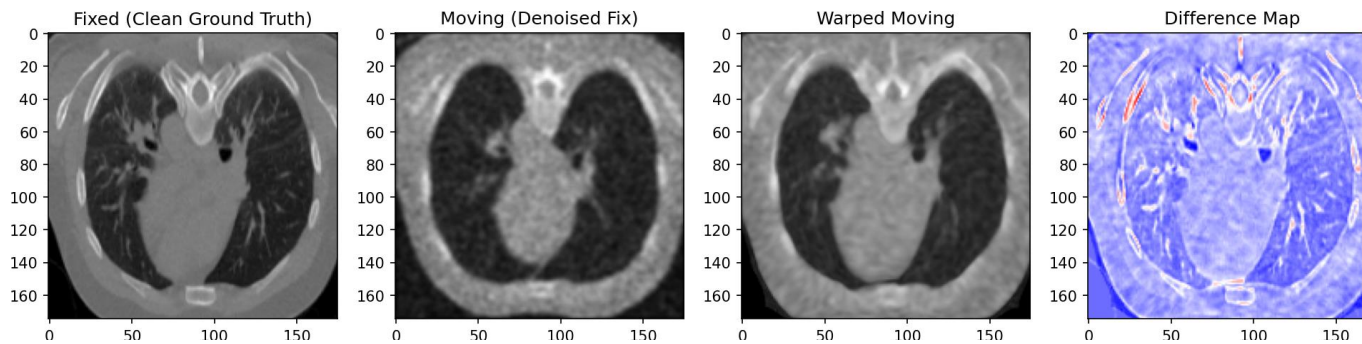


Corruption (noise)

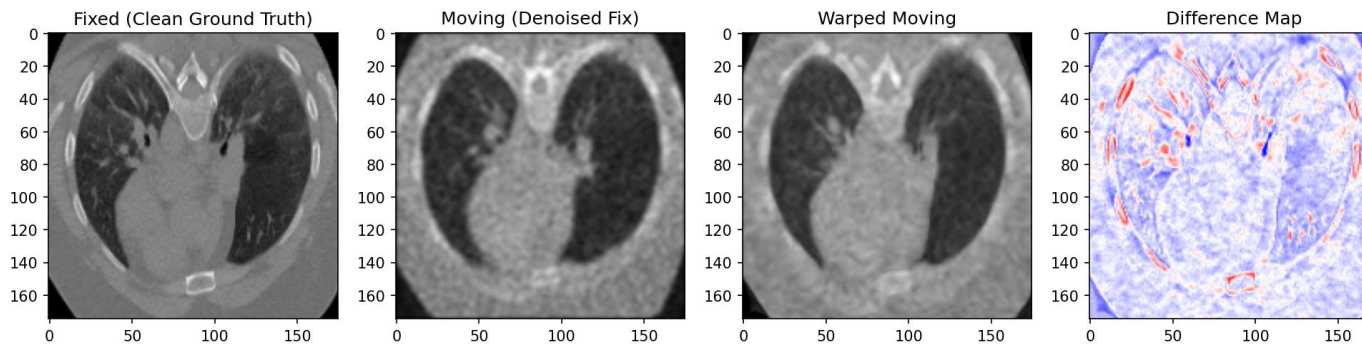
Pair 0001 — Fixed (Gaussian Signal Smoothing)



Pair 0002 — Fixed (Gaussian Signal Smoothing)

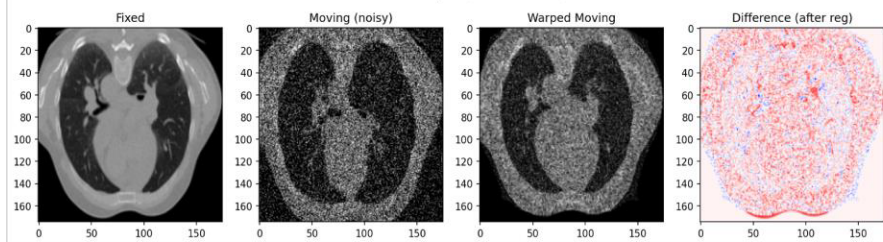


Pair 0003 — Fixed (Gaussian Signal Smoothing)

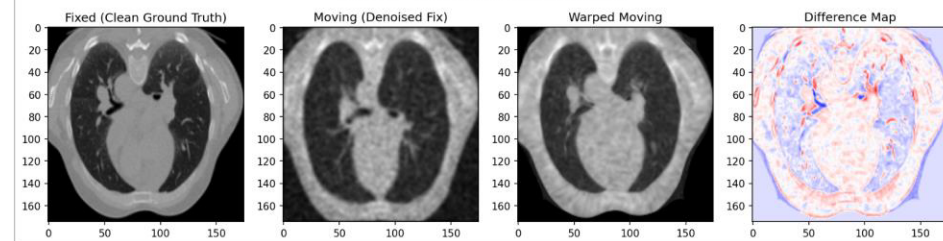


Corruption (noise)

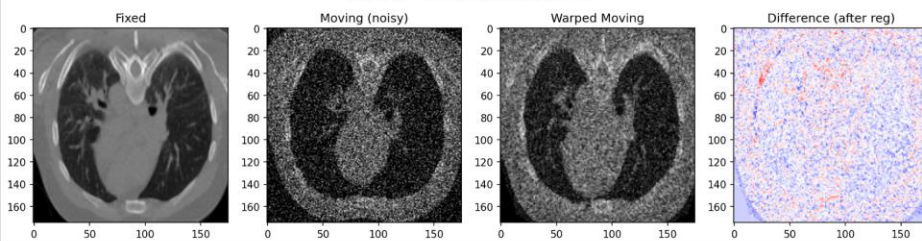
Pair 0001 — Corrupted (Gaussian Noise)



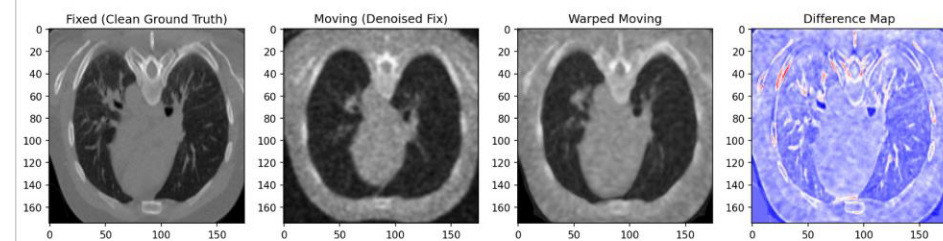
Pair 0001 — Fixed (Gaussian Signal Smoothing)



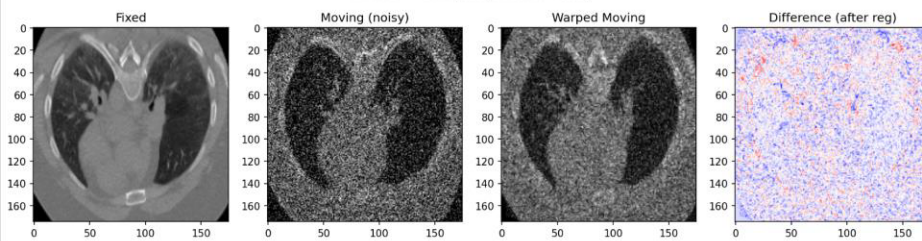
Pair 0002 — Corrupted (Gaussian Noise)



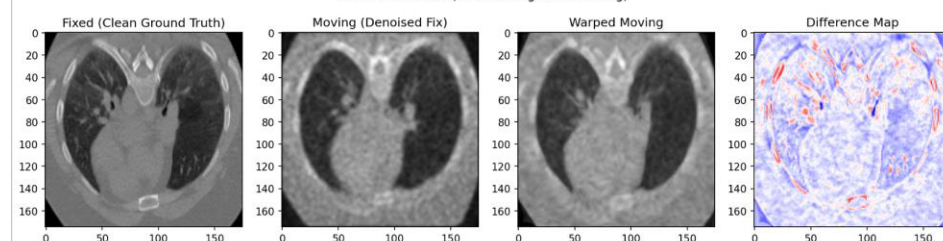
Pair 0002 — Fixed (Gaussian Signal Smoothing)



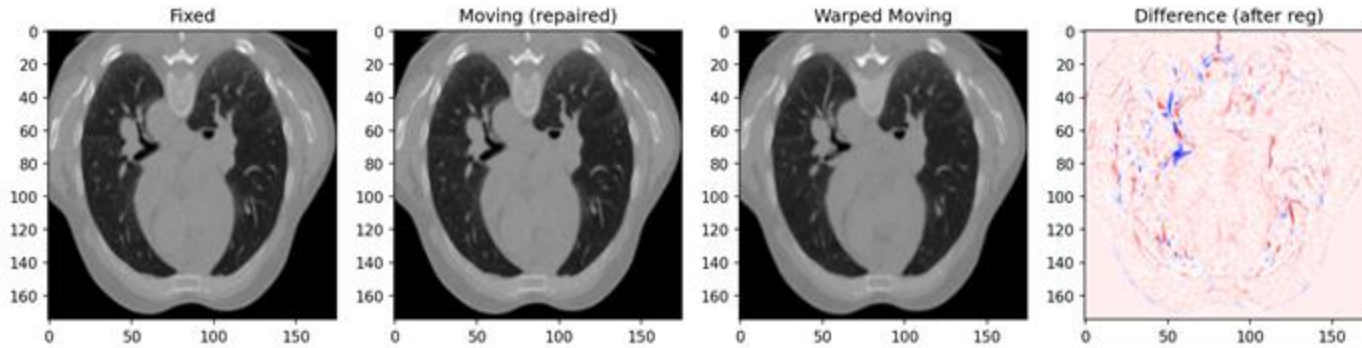
Pair 0003 — Corrupted (Gaussian Noise)



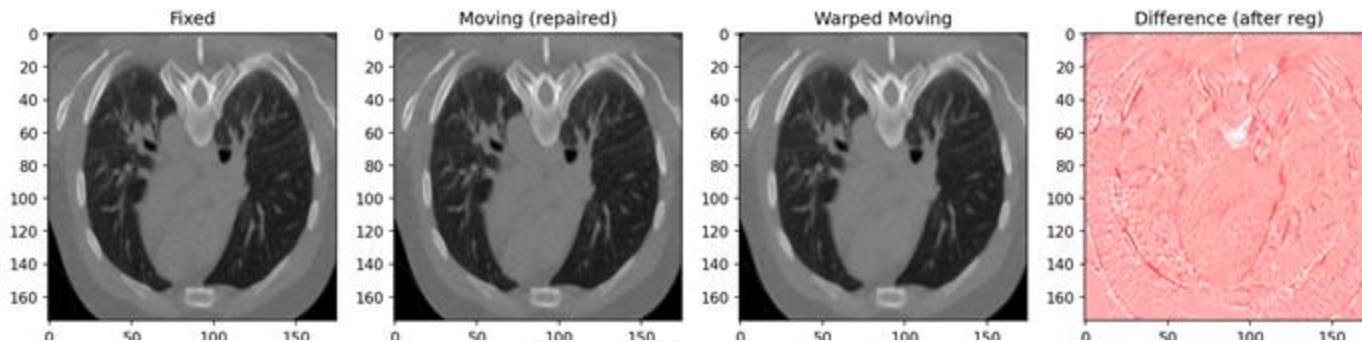
Pair 0003 — Fixed (Gaussian Signal Smoothing)



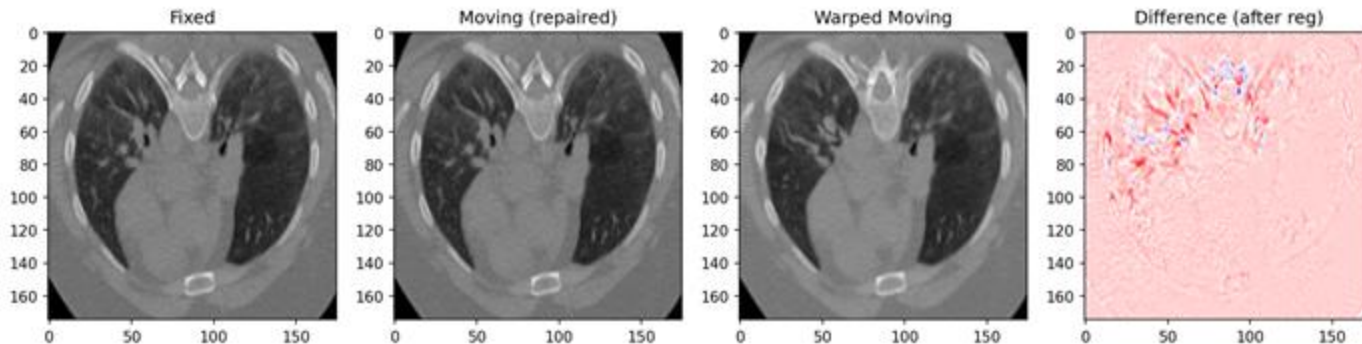
Pair 0001 — Fixed (Reference-based Inpainting)



Pair 0002 — Fixed (Reference-based Inpainting)

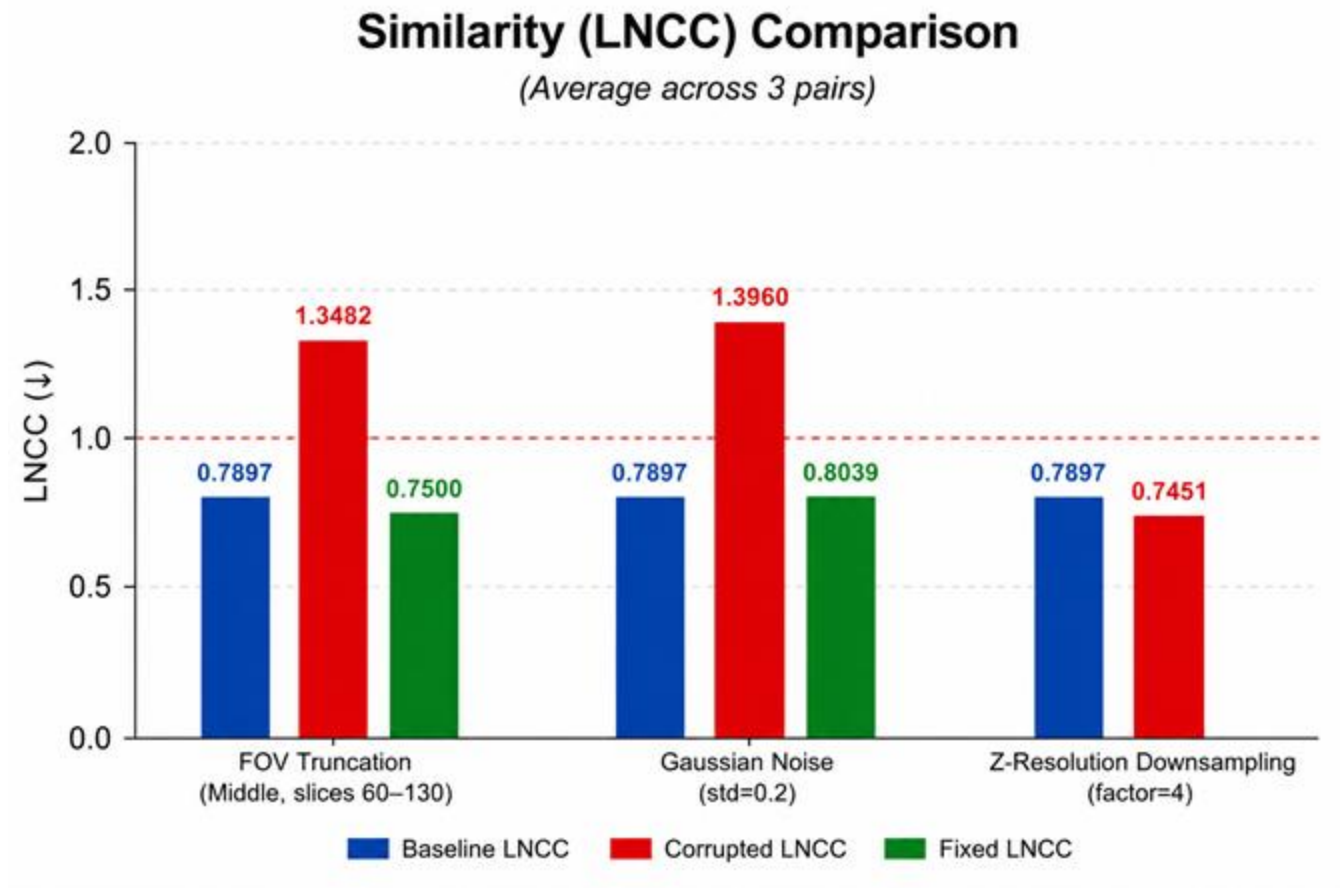


Pair 0003 — Fixed (Reference-based Inpainting)



Corruption (Reference based Inpainting)

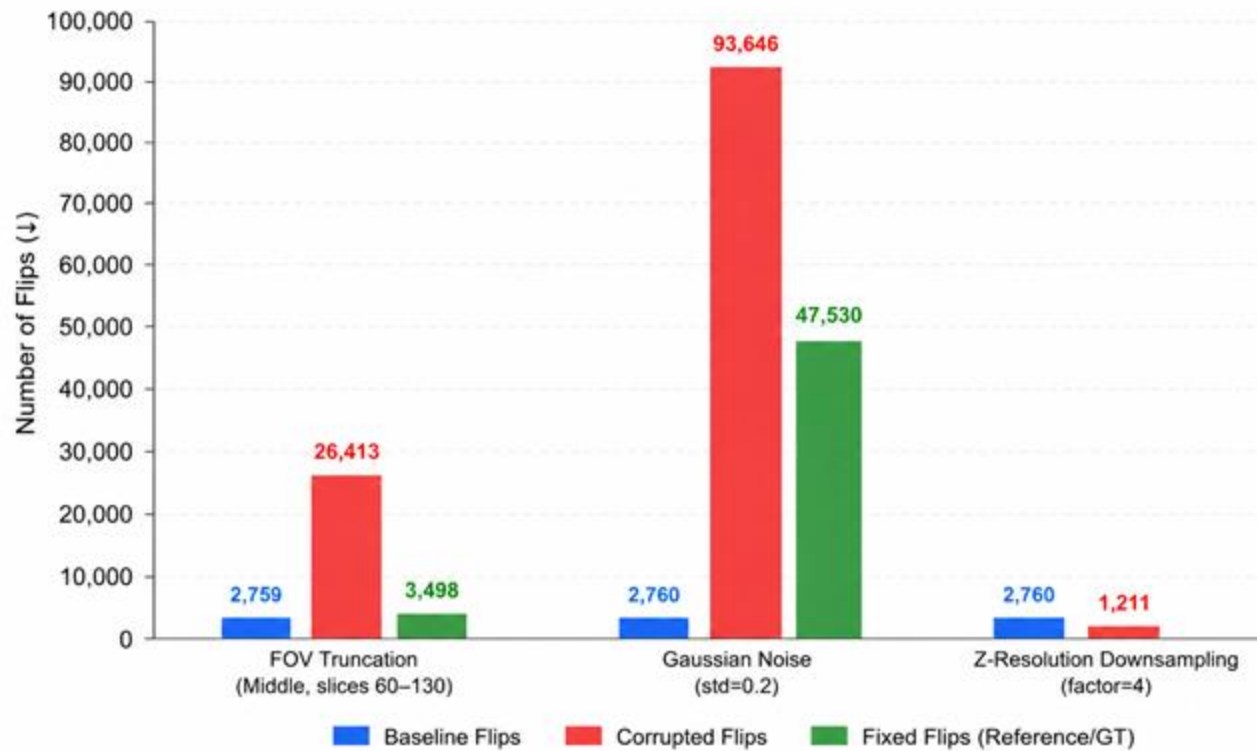
3- Results



3- Results

Flips (Jacobian Folding) Comparison

(Average across 3 pairs)



4- Summary

Model Vulnerability

uniGradICON reaches its limits under **extreme high-frequency noise**, resulting in heavily corrupted deformation fields

cannot compensate for **missing anatomical structures**

Downsampling

Reducing Z-resolution maintained or *slightly improved* registration

removing fine details, allowing the network to focus on **global lung shapes**

Successful Mitigation

Performance was restored through the implementation of Reference-based Inpainting



